

Study Guide

CHM 1045C Exam 1

Betancourt

Prerequisite Science Knowledge and Skills

1. Measurements and Conversions
2. Significant Digits, figures
3. Rounding
4. Significant Figures in calculations
5. Scientific Notation
6. Proper use of the calculator

The Metric System

1. Basic metric units
2. Prefixes
3. Conversions: metric – metric, metric – English
4. Volume and Density determinations
5. Temperature conversions

The Metric System

TABLE 1.4 Metric Conversions

Prefix	Factor	Symbol
giga–	10^9	G
mega–	10^6	M
kilo–	10^3	k
deci–	10^{-1}	d
centi–	10^{-2}	c
milli–	10^{-3}	m
micro–	10^{-6}	μ
nano–	10^{-9}	n
pico–	10^{-12}	p

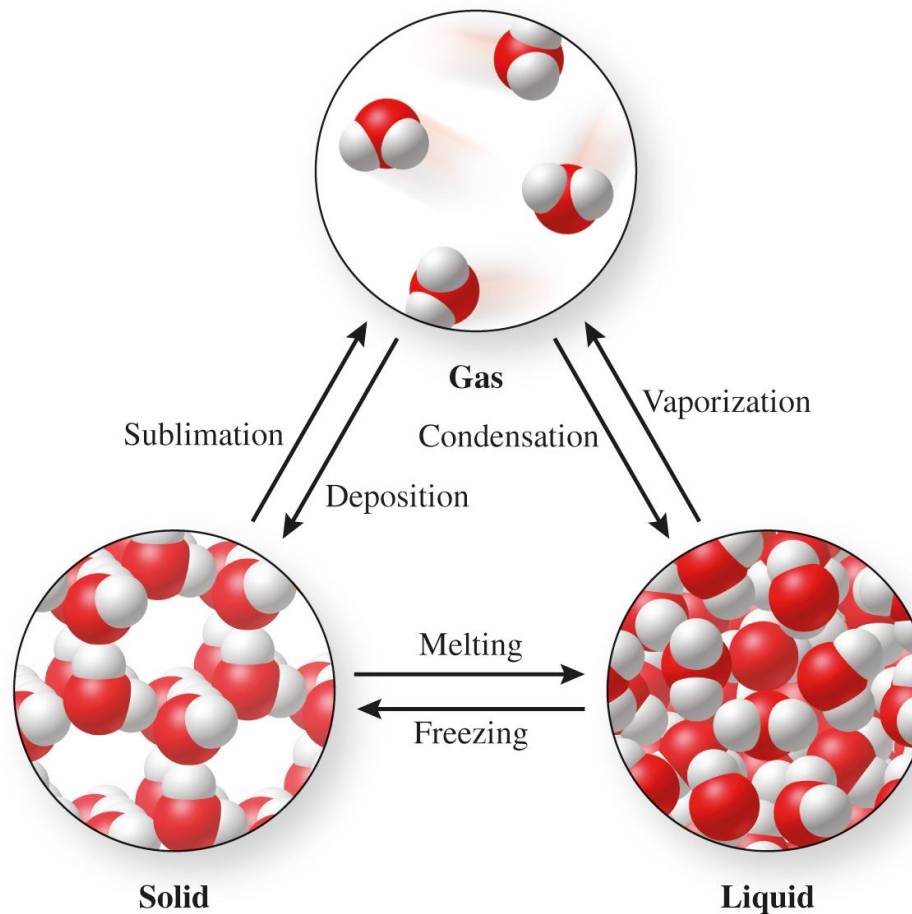
Matter and Energy

Study lecture notes and Chapter 1 in textbook

1. States of Matter and transitions
2. Matter: elements, compounds
3. Mixtures: homogeneous vs heterogeneous
4. Names and symbols of the elements
5. Physical vs Chemical changes

Matter and Energy

Memorize physical state transition shown



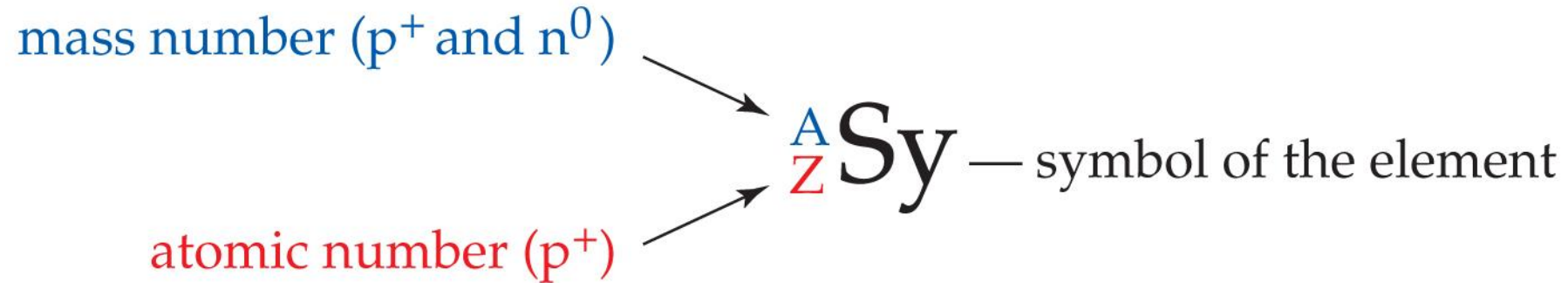
Model of Atom, Ions and Periodic Table

Study lecture notes and practice from Chapter 2

1. Dalton's Theory – postulates
2. Model of the atom : protons, neutrons and electrons
 - a. Cathode Ray Experiment
 - b. Gold Foil Experiment
3. Atomic Notation
 - isotopes
 - ions
 - determine number of protons, neutrons, electrons
4. Atomic Mass – determine average atomic mass

Model of Atom, Ions and Periodic Table

Focus on Atomic Notation, Atomic Number, Atomic Mass



Naming Compounds

Focus on Naming ionic binary and polyatomic compounds and molecular compounds (Chapter 2)

- Naming binary compounds
 - Naming binary compounds made from transition metals
 - Naming compounds with polyatomic ions
 - Naming covalent compounds
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- Empirical Formulas
 - Molecular formulas

